Design and Analysis of Compact BCD Adder for Future High-Speed and Low-Power Electronic Applications

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***Abstract*— In this study, a unique design of BCD adder optimized for area and speed is proposed. The design is assessed in comparison to current techniques, with an emphasis on important key metrics such as silicon area, power utilization, and propagation latency. The 16nm, 22nm, and 32nm technologies are used to model the circuits like BCD-MFA, BCD-SCLAA and BCD-DCLAA, and the results are correlated among the circuits respectively. When the operating temperature is varied from 27°C to 107°C, the T_d (delay) changes by 32.48%, 31.25% and 31.11% for the BCD-DCLAA implemented of 16 nm, 22 nm and 32 nm, respectively. The 4-bit BCD-MFA, BCD-SCLAA and BCD-DCLAA implemented with 16 nm CMOS exhibited a PDP of 5.56/6.64 fJ, 8.1417/10.386 fJ and 6.37/7.771 fJ, respectively at 27^0C/107^0C . Therefore, based on the examination of 3 different BCD adders considering latency, power and the product of T_d and P_c, it is observed that dynamic-CLAA BCDA is more efficient and accurate. Digital watches, digital meters, digital calculators, financial systems, microcontrollers, embedded systems, and data conversion systems that require accurate decimal arithmetic all use BCD adders.**

***Keywords*— Binary Coded Decimal Adder-BCDA, Mirror Full Adder-MFA, Static Carry look Ahead-SCLA, Dynamic Carry look Ahead-DCLA, Power delay product-PDP.**

I. INTRODUCTION

This BCD is widely employed in computers for encoding the decimal numbers [1]. BCDAs serve the purpose for the arithmetic operations utilizing binary representations of decimal values. BCD is an algorithm for expressing numbers that spans between 0 to 9 which employs 4-bit portrayals for every decimal digit [2]. BCDAs are essential for numerous modern digital systems [3], including accounting & financial systems, barcode scanners & UPC codes, automotive control units for fuel injection and medical devices & instruments, devices like glucose meters and infusion pumps etc [1][4]. Tingting zhang et al [5] used decimal group propagate and decimal group generate signals in order to determine carries in the BCDA. Hardware enactment often modify decimal fixed-point operands using BCD rather than binary [6][7]. Luigi dadda et al [8] recommend a technique to utilize a network of carry-free adders to compute the sum for every decimal column and then swiftly transform the sum into decimal format employing a well-known BCD converter.

Digital systems would struggle to produce precise decimal in case BCDAs weren't present [9]. The consequences comprise loss of decimal precision, error propagation, display issues and unreliability etc. Aspects to take into account while utilizing BCDAs are binary addition rules, valid BCD results, correction for invalid BCDAs [10]. Key components that are crucial for BCDAs are handling correction, preventing overflow, speed & efficiency, and testing & verification. Modern circuits are exploiting the nano technologies includes nano magnet logic as well as quantum-dot cellular automata and latest electronics gadgets are specifically 2D polarity-controllable transistors and spin-wave devices [3]. Novel developments & breakthroughs in BCDAs are reversible logic implementation, carry skip BCDAs, carry look ahead (CLA) BCDA etc [11]. CMOS circuits are being increasingly utilized within every aspect of modern electronics, because of their remarkable capabilities. The aforementioned circuits' unique attributes have led to an extensive spectrum of implementations throughout several industries. CMOS technology has portended significant advancements in electronics, facilitating high-performance, energy-efficient components like microprocessors and memory modules. Symmetrical CMOS logic circuits provide novel prospects for design creativity [12].

II. 4-BIT MIRROR FULL ADDER

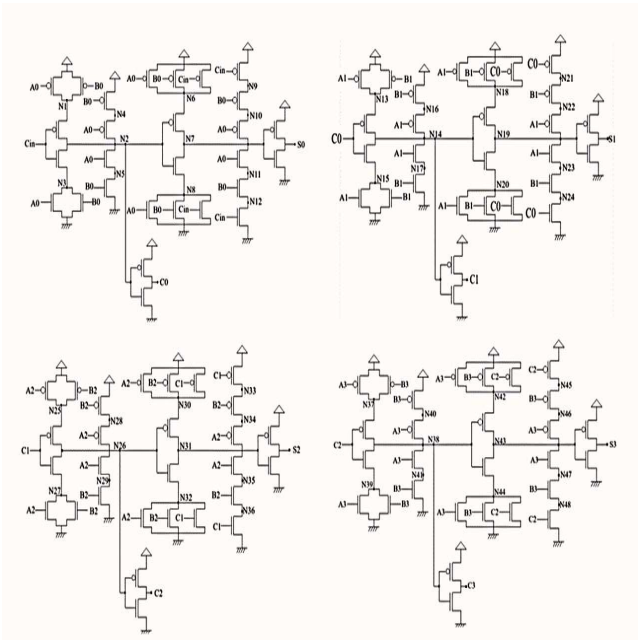


Fig. 1 4-Bit Mirror Full Adder

The 4-bit mirror full adder which is also denoted as 4-bit MFA, it comprises of four 1-bit MFA's. In 4-bit MFA, we have three input's A_i , B_i, C_{i-1}. Where C_{i-1} is carry in to the MFA, and two outputs sum (S_i) along with carry out (C_i). Each

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MFA generates sum and carry out. Precise BCD portrayal is ensured by the sum, and when modification is required, it is perceived by the carry-out. C_i from one MFA will be given as an input C_{i-1} to the successive MFA. Example, C_0 is carry out of first 1-bit MFA, which is used as input, as C_0 be the carry in for second MFA circuit.

III. BCDA USING 4-BIT MIRROR FULL ADDER

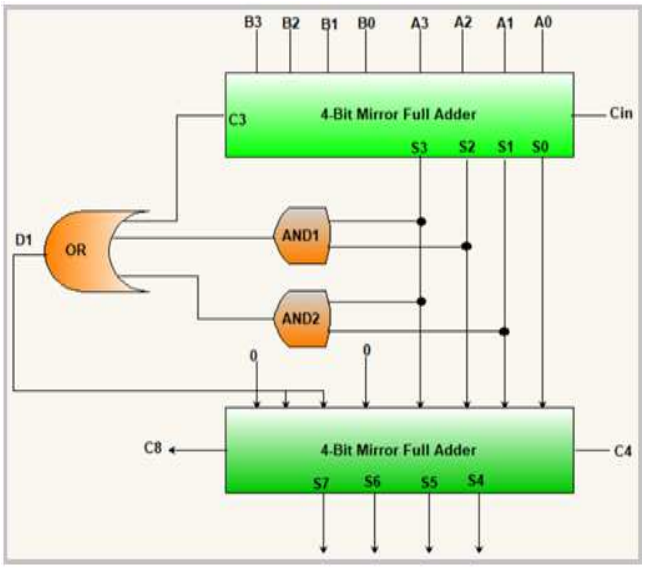


Fig. 2 Architecture diagram of BCD-MFA

Fig. 2. illustrates BCDA using 4-bit MFA, above BCD is implemented by two 4-bit MFAs, one OR gate & two AND gates. By adding two decimal digits we perform binary addition among the given inputs A_i, B_i and C_{in} until the outcome is in the range of 0 to 9. If the result extend binary number 9 then the result is invalid as per BCD algorithm. Therefore, we add binary number 6 (0110 in binary) for the correction of decimal number and the carry-out of each stage (barring the last phase) to the binary sum which helps propagate this correction.

The working of BCDA, which is mentioned in the figure 2. In BCD-MFA, A_i, B_i and C_{in} are the inputs and the outputs are sum (S_i) and C_i where i = 0,1,2,3. If the binary sum exceeds the range 9 then, the sum from 1st 4-bit binary will become input to the AND gates, here S_2, S_3 are the inputs to the AND1 & S_1, S_3 are the input labels for the 2nd AND gate. Where the outputs for AND1, AND2 and carry out C_3 from the upper 4-bit MFA are the inputs for the OR gate. Along with the output of OR gate D_1 and the binary sum of primary 4-bit MFA (S_0, S_1, S_2, S_3) are feed as input to the secondary 4-bit MFA. It is designed in such a way that the value coming out from D_1 is always '1' and D_0 , D_2 are given '0', which makes them appear like (0110) = 6.

Assign the values for input A_i = 5 (0101), B_i = 7 (0111) and C_{i-1} = '0' respectively. So, the resulting output from the first 4bit-MFA is C_i = 0 and S_i = 12 (1100) which is invalid result. By adding 6 (0110) to the sum which we are getting from the 4bit-MFA(criteria of BCD). Then the resulting output from the BCDA is C_i = 1 (0001) and where S_i = 2. It displays the resultant output be 12, which is valid result for the inputs 5 and 7.

IV. BCDA USING SCLAA

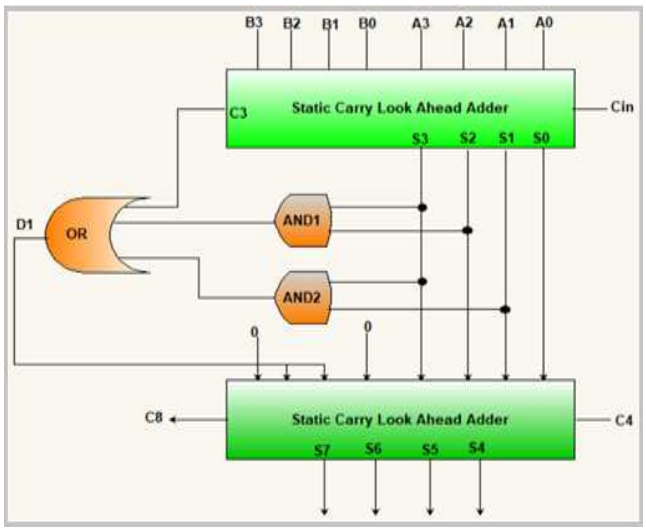


Fig. 3 Architecture diagram of BCD-SCLAA

The work in Fig. 3, presents the static carry look ahead adder(SCLAA). The carry signals are calculated during the entire operation, regardless of the clock signal. The SCLAA is easier to design & understand. Fig 4 illustrates the sample circuit of the SCLAA which consists circuit of propagate signal (Pi) and generate signal (Gi), two signals, were introduced [2] specifically to reduce the time required to generate the carry bits in BCDA. Fig. 4(a) depicts sample circuit of generate signal(Gi) and the Fig. 4(b) illustrates the propagate signal(Pi). Where expression for generative signal Gi=Ai. Bi & expression for propagate signal Pi = Ai $\oplus$ Bi where Ai & Bi are the inputs of Pi & Gi. (where i = 0,1,2,..7) here both the Gi and Pi are connected to both sample sum circuit as well as Carry circuit as the inputs which can be seen in Fig. 4(c) & Fig. (d) .

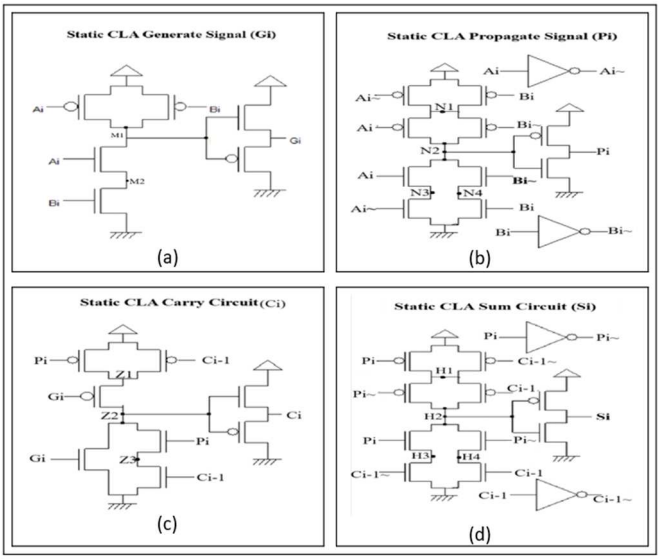


Fig. 4 Circuit schematic's of SCLAA

Fig. 4(c) demonstrates the circuit schematics of static carry look ahead adder, as carry (Ci) which is generated by the inputs Gi, Pi & Ci-1. Likewise, Ci-1 is the carry-in, it is one of the input bit. The transistor Pmos with date bits Pi & Ci-1 are parallel as well as Nmos with data bits Pi and Ci-1 are in series which result in multiplication of those two inputs and

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transfers the result. Transistor Pmos with input Gi is in series in addition with Nmos transistors input Gi is in parallel with Pi and Ci-1 which results to addition of inputs, which expressed as carry out is Ci = Gi + Pi. Ci-1.

V. BCDA USING DCLAA

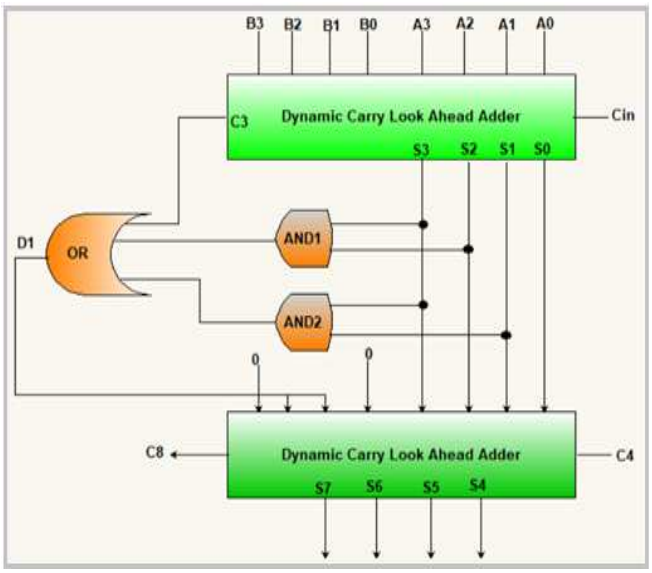


Fig. 5 Architecture diagram for BCD-DCLAA

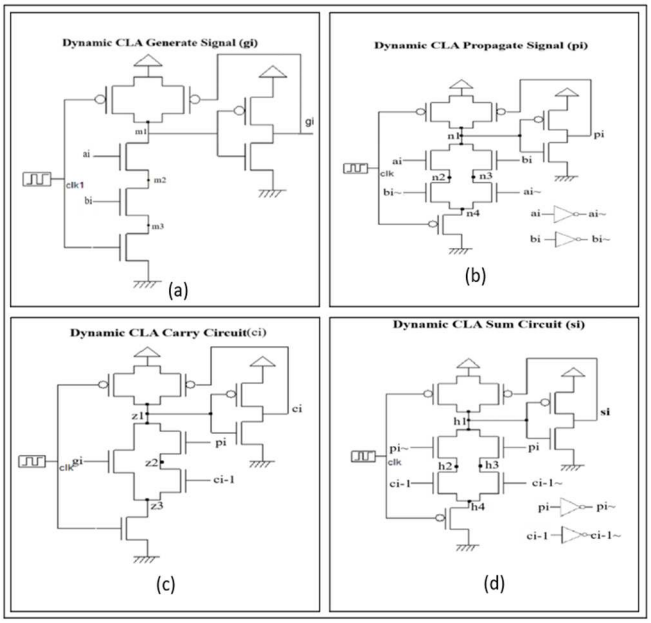


Fig. 6 Circuit schematic of DCLAA

As shown in Fig 5, BCDA is implemented by using DCLAA to enhance performance, focusing on both speed and reliability. By minimizing the time required to figure out carry bits, the carry look ahead adder enhances the accuracy and efficiency. CLAA computes carry signals much earlier, compared to ripple carry adder - RCA, which hold off for carry signals to propagate. CLAA speeds up both addition as well as remaining arithmetic operations by lowering the carry transmission latency. Two perspectives are the foundation of the CLA adder logic's are carry generate(G_i) and carry propagate(P_i). In DCLAA there are 2 phases one is pre-charge phase and another evaluate phase. At first the DCLAA will be in pre-charge phase, no carry will be generated.

The carry signals are computed when needed (evaluation phase). One of the speciality of the DCLAA is having dynamic node, which can also work as switch between pre-charge and evaluate phase. m1 is the Dynamic node in Fig.6(a), n1,z1 and h1 are dynamic node in Fig. 6(b), 6(c), and 6(d) respectively. In DCLAA the transistors are used efficiently, but complex to design while compared to SCLAA, because additional clocking required. Dynamic logic relies on clock signal for proper operation. It controls the pre-charge and evaluate phase. The generate signal(gi) and propagate signal(pi) plays a crucial role. In Fig.6(a) in generate signal (gi) sample circuit, we can analyse that irrespective of the carry-in ci-1, the gi demonstrates, whether a carry is generated during both the input bits ai and bi are 1, where ai, bi and ci-1 (i=0,1,2...7) are input bits to the DCLAA. In Fig 10, we have added the clock pulse as one of the input values and gi, pi, si and ci are given as a feedback to PMOS transistor to bring out the remaining leakage power supply from the transitions, which gives fastest results.

VI. SHORT CIRCUIT POWER ANALYSIS

Short circuit (SC) power represents the amount of energy dissipation from the circuits during signal transitions which leads to device heat up and consumes more power from the input voltage in order to meet the target . Fig 7 illustrates the analysis of short circuit power at different gate lengths (L_G) for the BCD adders using different adders as aforementioned. Short circuit power of BCD adder using MFA at the gate length of 16 nm for the temperature ranging from 27°C/107°C is approx. 128µW/230µW and the SC power is increased by 44.35% which is comparatively higher than dynamic CLA. Likewise, the SC power of BCD-MFA adder at the gate length of 22 nm technology for temperature varies from 27°C to 107°C is approximately 210µW to 390µW and the increase is 46.16%. Nevertheless, SC power of BCD-MFA at the gate length of 32 nm technology for temperature from 27°C to 107°C is in the vicinity of 420µW to 760µW. SC power of BCD-A using SCLAA circuit for the gate length of 16 nm for the temperature ranging from 27°C/107°C is nearly 160µW to 270µW. SC power of BCD-SCLA at the gate lengths of 22 nm technology for temperature ranging from 27°C-107°C is in the vicinity 260µW/470µW. SC power of BCD-SCLAA for the gate length of 22 nm technology for the temperature varying from 27°C to 107°C is approximately 510µW to 840µW.

SC power of BCD-A using DCLAA at the gate length of 16 nm for 27°C/107°C is around 110µW / 170µW. SC power of BCD-DCLAA at the gate lengths of 22 nm for 27°C to 107°C is around 180µW to 320µW. SC power of BCD-DCLAA at the gate lengths of 22 nm technology for the temperature ranging from 27°C to 107°C is around 280 µW to 480 µW. The above results implies that the SCLAA used BCDA consumes more power than MFA based BCD adder. Also, the MFA based BCD adder has low power consumption compared to BCD-DCLA. Compared to 4-bit BCD-MFA and BCD-SCLAA, 4-bit BCD-DCLAA implemented using 16 nm CMOS with 0.7 V V_{DD} offer 14.1/26.1 % and 31.3/37.1 % reduction in SC power at 27°C/1077°C. Similarly, compared to 4-bit BCD-MFA and BCD-SCLAA, 4-bit BCD-DCLAA implemented using 32 nm CMOS offer 33.4/45.1 % and 36.9/42.9 % reduction in SC power at 27°C/1077°C.

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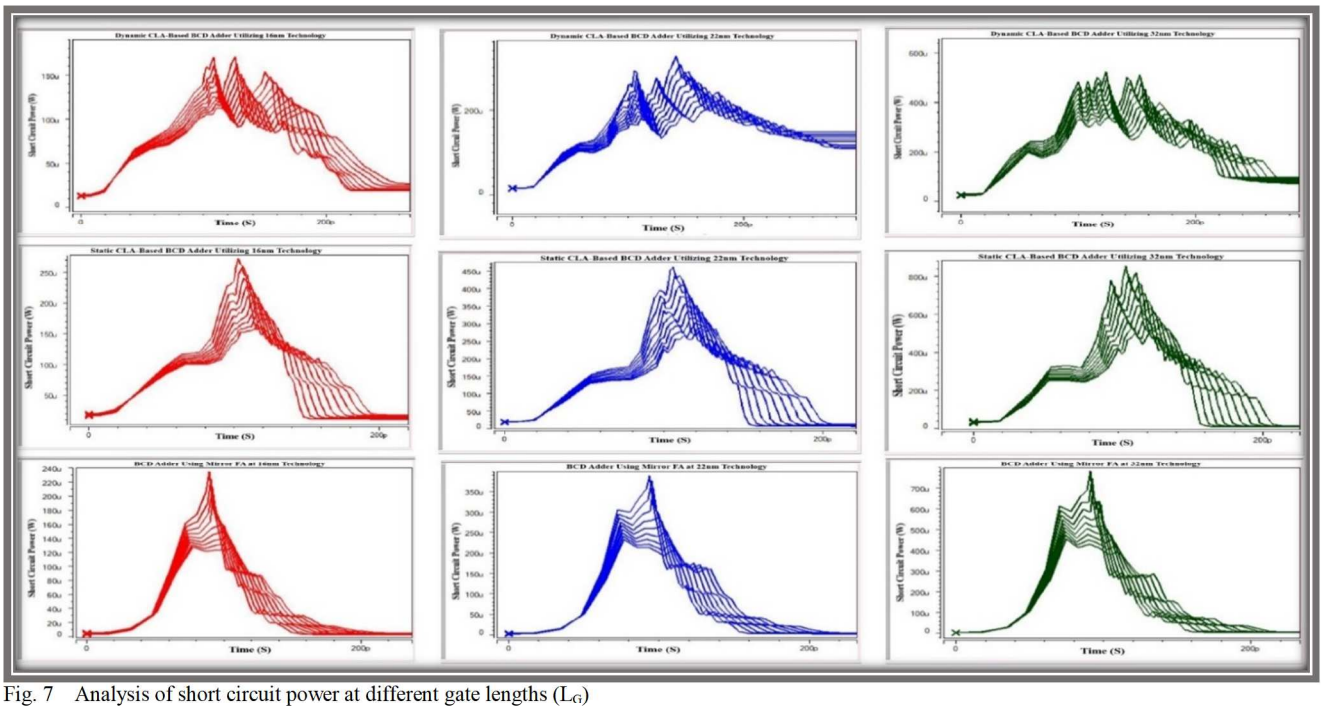


Fig. 7 Analysis of short circuit power at different gate lengths (L_G)

VII. DELAY & POWER ANALYSIS

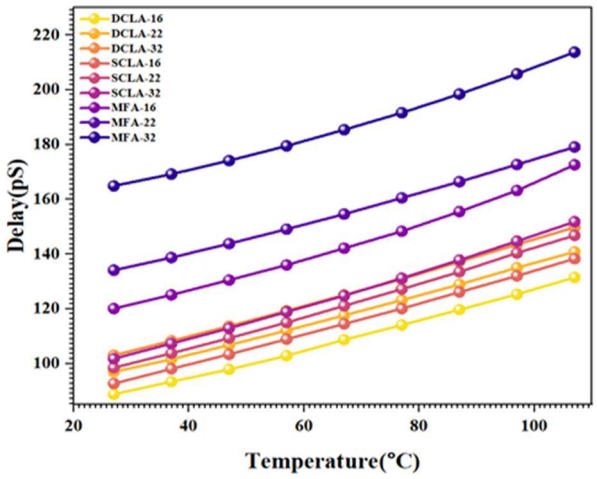


Fig. 8 Differentiation of latency for 3 technologies

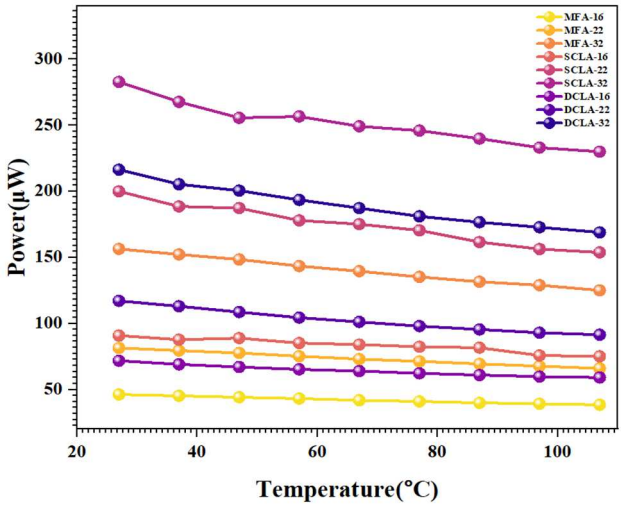


Fig. 9 Differentiation of power for 3 technologies

The 4-bit BCD-MFA implemented with 16 nm CMOS and a supply of 0.7 V offered a T_d of 120.1 pS and 172.6 pS at 27^0C and 107^0C, respectively. Similarly, the 4-bit BCD-MFA implemented with 22 nm CMOS and a supply of 0.8 V offered a T_d of 134.1 pS and 179.1 pS at 27^0C and 107^0C, respectively. Finally, the 4-bit BCD-MFA implemented with 32 nm CMOS and a supply of 0.9 V offered a T_d of 164.9 pS and 213.7 pS at 27^0C and 107^0C, respectively. The 4-bit BCD-SCLAA implemented with 16 nm CMOS and a supply of 0.7 V offered a T_d of 92.7 pS and 138.3 pS at 27^0C and 107^0C, respectively. Similarly, the 4-bit BCD-SCLAA implemented with 22 nm CMOS and a supply of 0.8 V offered a T_d of 98.5 pS and 146.8 pS at 27^0C and 107^0C, respectively. Finally, the 4-bit BCD-SCLAA implemented with 32 nm CMOS and a supply of 0.9 V offered a T_d of 101.8 pS and 151.8 pS at 27^0C and 107^0C, respectively. Likewise, the 4-bit BCD-DCLAA implemented with 16 nm CMOS and a supply of 0.7 V offered a T_d of 88.8 pS and 131.5 pS at 27^0C and 107^0C, respectively. Similarly, the 4-bit BCD-DCLAA implemented with 22 nm CMOS and a supply of 0.8 V offered a T_d of 97 pS and 140.8 pS at 27^0C and 107^0C, respectively. Finally, the 4-bit BCD-DCLAA implemented with 32 nm CMOS and a supply of 0.9 V offered a T_d of 103 pS and 149.8 pS at 27^0C and 107^0C, respectively (Fig. 8).

The 4-bit BCD-MFA implemented with 16 nm CMOS and a supply of 0.7 V consumed a power of 46.3 µW and 38.5 µW at 27^0C and 107^0C, respectively. Similarly, the 4-bit BCD-MFA implemented with 22 nm CMOS and a supply of 0.8 V consumed a power of 81.5 µW and 66 µW at 27^0C and 107^0C, respectively. Finally, the 4-bit BCD-MFA implemented with 32 nm CMOS and a supply of 0.9 V consumed a power of 156.4 µW and 125 µW at 27^0C and 107^0C, respectively. The 4-bit BCD-SCLAA implemented with 16 nm CMOS and a supply of 0.7 V consumed a power of 90.8 µW and 75.1 µW at 27^0C and 107^0C, respectively. Similarly, the 4-bit BCD-SCLAA implemented with 22 nm CMOS and a supply of 0.8 V consumed a power of 199.9 µW and 153.7 µW at 27^0C and 107^0C, respectively. Finally, the 4-bit BCD-SCLAA implemented with 32 nm CMOS and a supply of 0.9 V

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consumed a power of 282.7 µW and 229.9 µW at 27^{0}C and 107^{0}C, respectively. Likewise, the 4-bit BCD-DCLAA implemented with 16 nm CMOS and a supply of 0.7 V consumed a power of 71.8 µW and 59.1 µW at 27^{0}C and 107^{0}C, respectively. Similarly, the 4-bit BCD-DCLAA implemented with 22 nm CMOS and a supply of 0.8 V consumed a power of 117.1 µW and 91.4 µW at 27^{0}C and 107^{0}C, respectively. Finally, the 4-bit BCD-DCLAA implemented with 32 nm CMOS and a supply of 0.9 V consumed a power of 216.3 µW and 168.9 µW at 27^{0}C and 107^{0}C, respectively (Fig. 9).

VIII. Power Delay Product Analysis

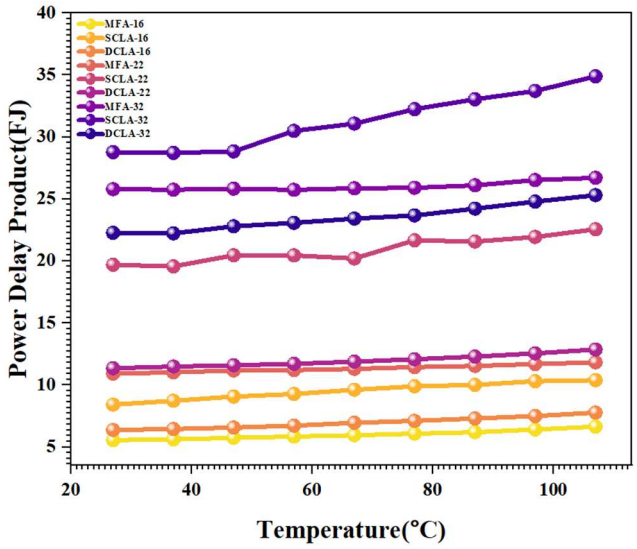


Fig. 10 PDP of 3 BCD circuits

TABLE I. PERFORMANCE OF BCD ADDERS

Ref.	4-Bit BCD Adder Performance Comparison		
	Technology (nm)	*Power (µW)*	*Delay (pS)*
4	45	49.5	500
4	180	1990	500
13	45	17.8	102000
13	45	0.5	98600
14	180	8430	NR
14	180	6250	NR
This work	16	46.3	120.1
	22	81.5	134.1
	32	156.4	164.9

In digital circuit design, the PDP measure is very crucial, especially in the case of semiconductors and low-power electronics. The technologies used to implement these works include 16nm, 22nm, and 32nm. The 4-bit BCD-MFA implemented with 16 nm CMOS and a supply of 0.7 V exhibited a PDP of 5.56 fJ and 6.64 fJ at 27^{0}C and 107^{0}C, respectively. Similarly, the 4-bit BCD-MFA implemented with 22 nm CMOS and a supply of 0.8 V exhibited a PDP of 10.90 fJ and 1.82 fJ at 27^{0}C and 107^{0}C, respectively. Finally, the 4-bit BCD-MFA implemented with 32 nm CMOS and a supply of 0.9 V exhibited a PDP of 25.79 fJ and 26.71 fJ at 27^{0}C and 107^{0}C, respectively. The 4-bit BCD-SCLAA implemented with 16 nm CMOS and a supply of 0.7 V exhibited a PDP of 8.41 fJ and 10.38 fJ at 27^{0}C and 107^{0}C, respectively. Similarly, the 4-bit BCD-SCLAA implemented with 22 nm CMOS and a supply of 0.8 V exhibited a PDP of 19.69 fJ and 22.56 fJ at 27^{0}C and 107^{0}C, respectively. Finally, the 4-bit BCD-SCLAA implemented with 32 nm CMOS and a supply of 0.9 V exhibited a PDP of 28.77 fJ and 34.89 fJ at 27^{0}C and 107^{0}C, respectively. Likewise, the 4-bit BCD-DCLAA implemented with 16 nm CMOS and a supply of 0.7 V exhibited a PDP of 6.375 fJ and 7.77 fJ at 27^{0}C and 107^{0}C, respectively. Similarly, the 4-bit BCD-DCLAA implemented with 22 nm CMOS and a supply of 0.8 V exhibited a PDP of 11.35 fJ and 12.86 fJ at 27^{0}C and 107^{0}C, respectively. Finally, the 4-bit BCD-DCLAA implemented with 32 nm CMOS and a supply of 0.9 V exhibited a PDP of 22.27 fJ and 25.3 fJ at 27^{0}C and 107^{0}C, respectively (Fig. 10). Table 1 presents the performance comparison of the advanced 4-bit BCD adders

IX. Conclusion

In this study, we used mirror FA, static CLA and dynamic CLA to study the performance of BCDA. The gate length(L_G), power consumption(P_c) and transmission delay(T_d) are the main focus of this study. We concluded in terms of T_d in different gate lengths the dynamic CLA based BCDA performs far better when compared to remaining. In point of power consumption the MFA based BCDA performs better. In of both the area of power utilization and delay. The static CLA based BCDA is performing average. BCD implemented by DCLA adder is preferable in high speed applications. MFA based BCD can use in less operating power devices. static CLA are simpler but slower, while dynamic CLA are faster but more complex. The choice depends on the trade-offs between speed, area, and power consumption in a specific application. The proposed BCD adder offers 76% improvement in speed compared with state of the art 4-bit BCD adder at low power as shown in table 1.

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